



Abd Alrahman Ahmed Mousa Mohamed

Computer Science · Software Developer

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Projects

1. Cosmic Observer

Windows astronomy desktop app — 3D sky, NASA data, AI tutor.

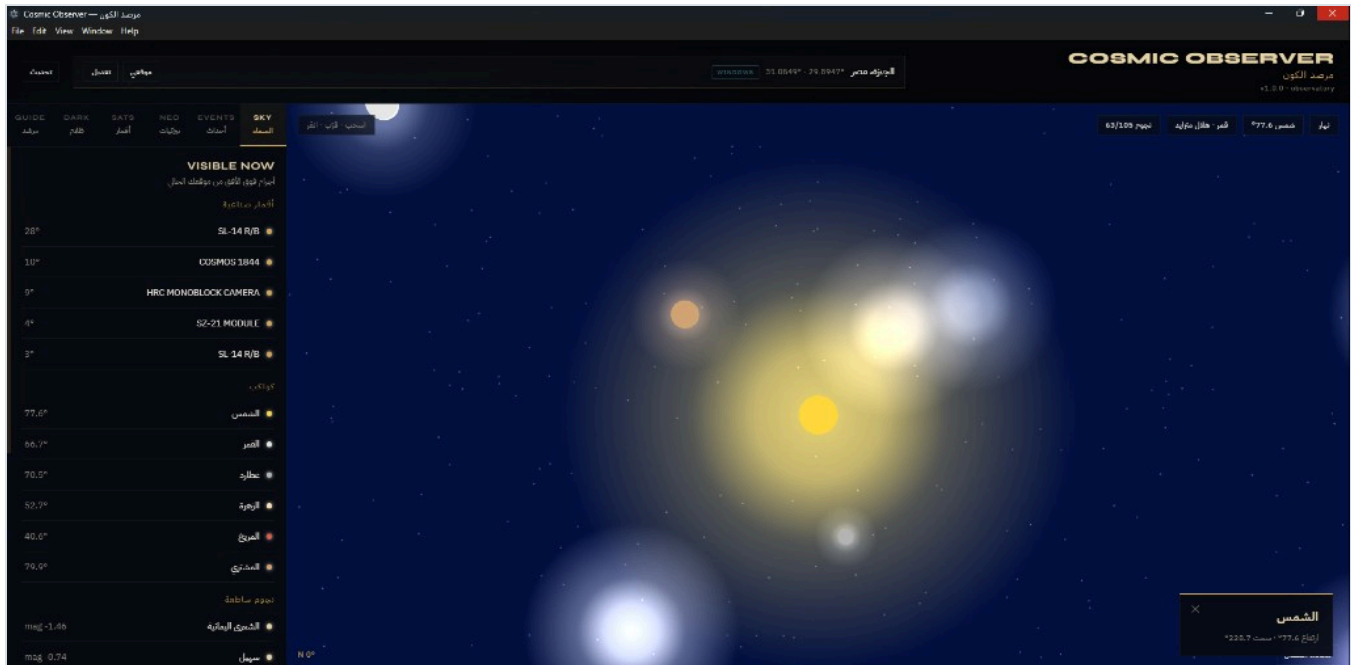
2. Vertical Farm Analysis

Python pipeline — farm telemetry to charts and PDF report.

Software developer focused on desktop applications and data analysis pipelines. Work is delivered with runnable demos, screenshots, and client-ready reports rather than slides alone.

Cosmic Observer

Windows desktop astronomy application with an interactive 3D sky, near-Earth object tracking, orbital debris monitoring, a dark-matter research module, and an AI astronomy tutor. Shipped as a native Windows installer and portable executable.



Capabilities

- Live 3D sky from GPS or manual location
- Satellites, planets, and bright stars list
- Sun / Moon status and sky events
- Modules: Sky, Events, NEO, Sats, Dark, Guide
- Arabic and English interface

Links

- github.com/engboda6/cosmic-observer
- License: MIT
- Platform: Windows (.exe)
- Demo video available on portfolio site

Cosmic Observer — Stack & Features

Built with JavaScript throughout: React and Three.js for the interface and sky rendering, Node.js/Express for local APIs and orbital calculations, Electron for desktop packaging.

Technologies

JavaScript	React 18	Three.js	Vite	Node.js	Express	astronomy-engine	OpenAI SDK	Electron 33
GLSL	PowerShell							

Features

- Interactive 3D celestial sphere
- Sunrise/sunset and moon phases
- NASA NeoWs near-Earth object tracking
- CelesTrak orbital debris / TLE
- Dark-matter lensing research views
- Optional OpenAI astronomy tutor

Data sources

- NASA NeoWs
- JPL Sentry / CAD
- NASA CNEOS Fireball
- CelesTrak (TLE)
- VizieR / CDS

Structure

- electron/ — desktop shell
- server/ — local APIs
- src/ — React UI + Three.js

Vertical Farm Telemetry Analysis

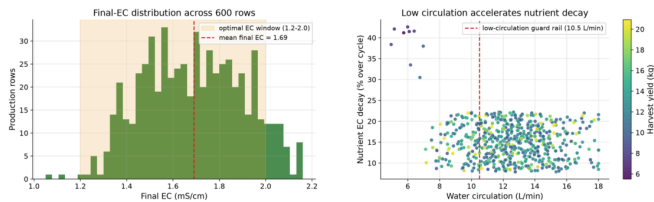
Pure Python analysis of 600 farm telemetry rows. Report prepared by **Abd Alrahman Ahmed**. Below: page 1 of the delivered PDF.

This report analyses 600 production rows of real agricultural automation telemetry and water-chemistry logs across the vertical farm (6 stacks x 10 racks x 10 rows = 60 irrigation channels). The objective is to map where nutrient-solution EC decays, where plant growth stalls, and how circulation, light-cycle hours and humidity drive vertical-stack yields and operating cost.

Key metrics

Production rows analysed	600
Irrigation channels (Stack-Rack)	60
Mean final EC	1.69 mS/cm
Rows within optimal EC window	92%
Mean EC decay per cycle	15.3%
Rows with stalled growth	8 (1%)
Low-circulation channels	12 of 60
HIGH-priority decay channels	12
Total harvest yield	7,923 kg

- Key findings:
1. Water circulation is the dominant lever: channels with low flow show markedly faster EC decay and a higher share of stalled growth.
 2. Faster nutrient decay drives EC away from the optimal window, which is a leading indicator of stalled growth and lost yield.
 3. Yield loss is not uniform - it clusters in a small set of channels that can be fixed with targeted maintenance rather than facility-wide change.
 4. Worst channel is D-R07 (score 0.88, 19% decay, 11.0 L/min).



Libraries, Outputs & Contact

The project is written entirely in Python and converts a raw farm CSV into analysis tables, visualizations, and a report suitable for client delivery.

Libraries

- **pandas** — data cleaning and aggregation
- **numpy** — numerical calculations
- **matplotlib** — charts and PDF figures
- **openpyxl** — Excel export

Deliverables

- Clean structured dataset
- Analysis across 60 irrigation channels
- Charts and Excel workbooks
- Three-page nutrient analysis PDF

Contact

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